Project **Mansourah Massarah Gold Project**

Customer: **MA’ADEN GOLD AND BASE METALS (MA’ADEN)**

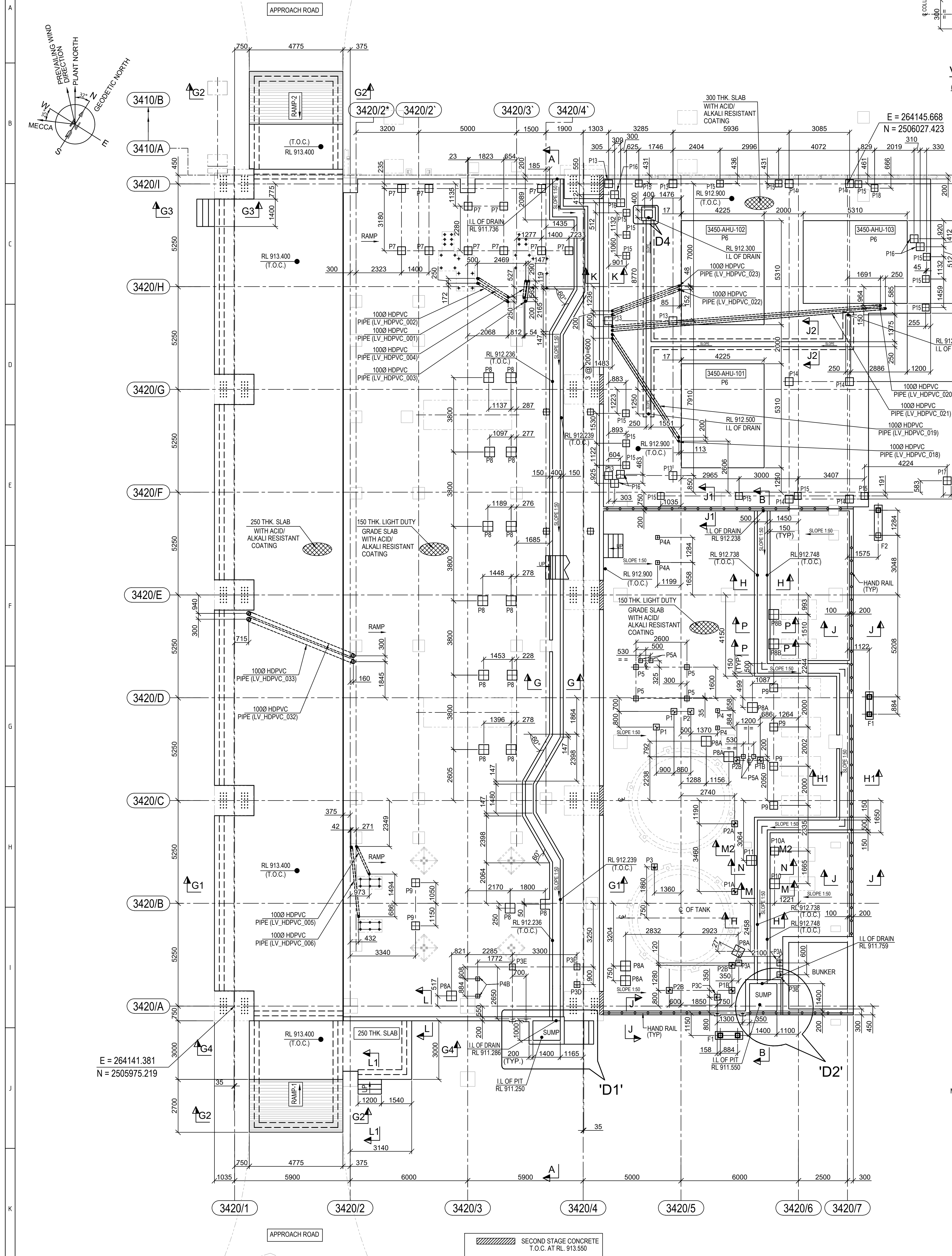
Title: **GENERAL ARRANGEMENT DRAWING PRESSURE OXIDATION- AUTOCLAVE BUILDING LAYOUT & DETAIL OF GRADE SLAB**

Document Number: **MD-216-3420-EG-CV-GAD-1245**
Revision: **D01**
Revision Date: **15.03.2023**
Status: **ASBUILT**

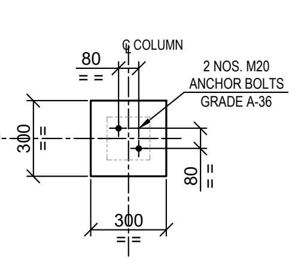
Outotec Document Number :	900707	GAU01	HH12	MTL01	34250	OU100704172	6
L&T Document Number :	O19045	450	C	G	24566	001	6

REVISION HISTORY

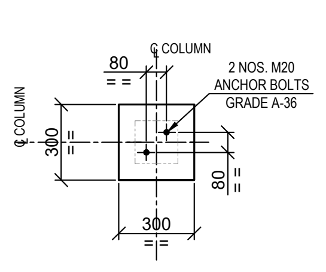
Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	20.10.2020	FIRST ISSUE	SOR	SJM	AG	
C01	21.11.2020	ISSUED FOR CONSTRUCTION	SOR	SJM	AG	
C02	04.01.2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C02	SOR	SJM	AG	
C03	25.01.2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C03	SOR	SJM	AG	
C04	26.02.2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C04	SOR	SJM	AG	
C04	15.03.2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C05	SOR	SJM	AG	
D01	15.03.2023	AS BUILT	SOR	SJM	AG	


APPROACH ROAD
PREVAILING WIND DIRECTION
PLANT NORTH
GEODETIC NORTH
MECCA
RAMP-2
RAMP-1
E = 264145.668
N = 2506027.423
E = 264141.381
N = 2505975.219
250 THK. SLAB WITH ACID/ ALKALI RESISTANT COATING
150 THK. LIGHT DUTY GRADE SLAB WITH ACID/ ALKALI RESISTANT COATING
300 THK. SLAB WITH ACID/ ALKALI RESISTANT COATING
SUMP
BUNKER
HAND RAIL (TYP)
'D1'
'D2'
SECOND STAGE CONCRETE T.O.C. AT RL 913.550

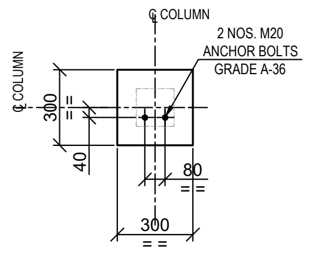
LAYOUT OF GRADE SLAB
(SCALE - 1:100)



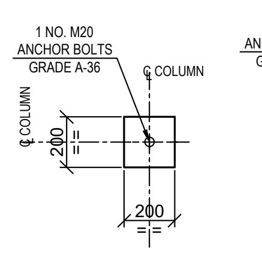
VIEW : F1-F1
(ONLY FOR P1,P1A,P1B)
SCALE 1:20



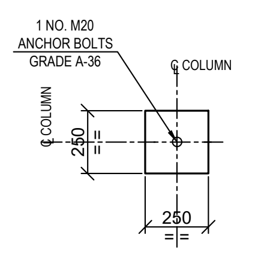
VIEW : F1-F1
(ONLY FOR P2,P2A,P2B)
SCALE 1:20



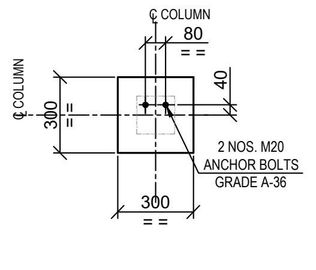
VIEW : F1-F1
(ONLY FOR P3, P3D)
SCALE 1:20



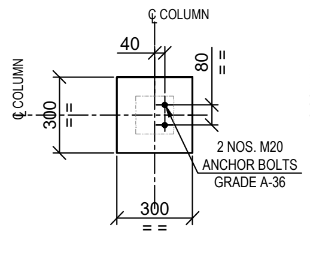
VIEW : F2-F2
(ONLY FOR P4, P4A & P4B)
SCALE 1:20



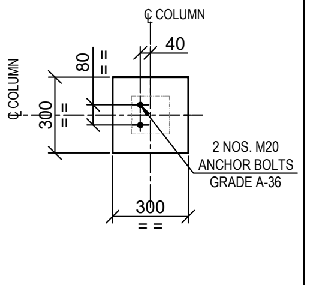
VIEW : F2-F2
(ONLY FOR P5)
SCALE 1:20



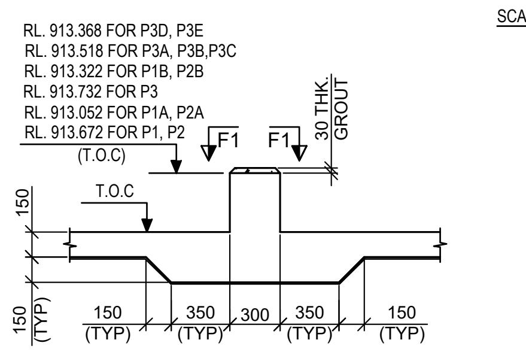
VIEW : F1-F1
(ONLY FOR P3A, 3E)
SCALE 1:20



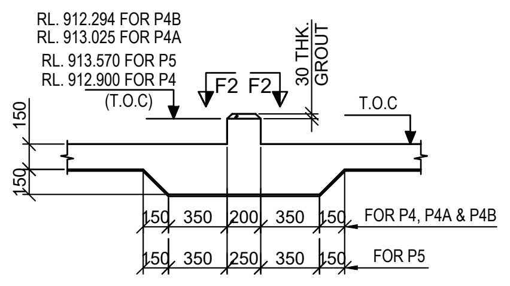
VIEW : F1-F1
(ONLY FOR P3B)
SCALE 1:20



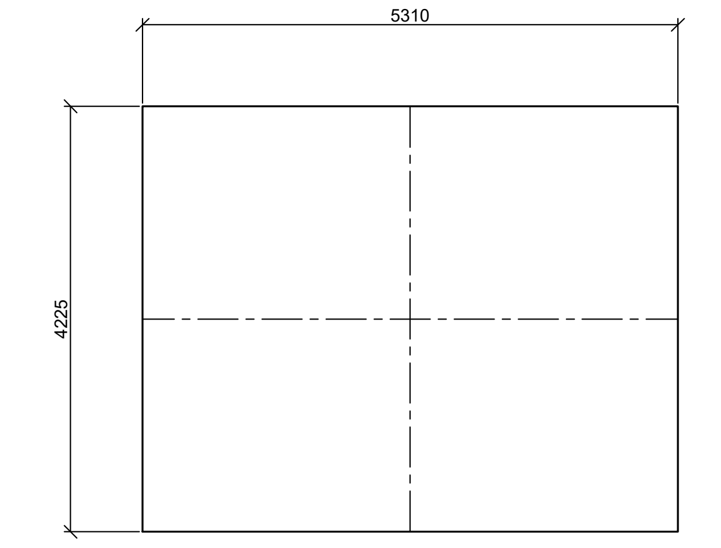
VIEW : F1-F1
(ONLY FOR P3C)
SCALE 1:20



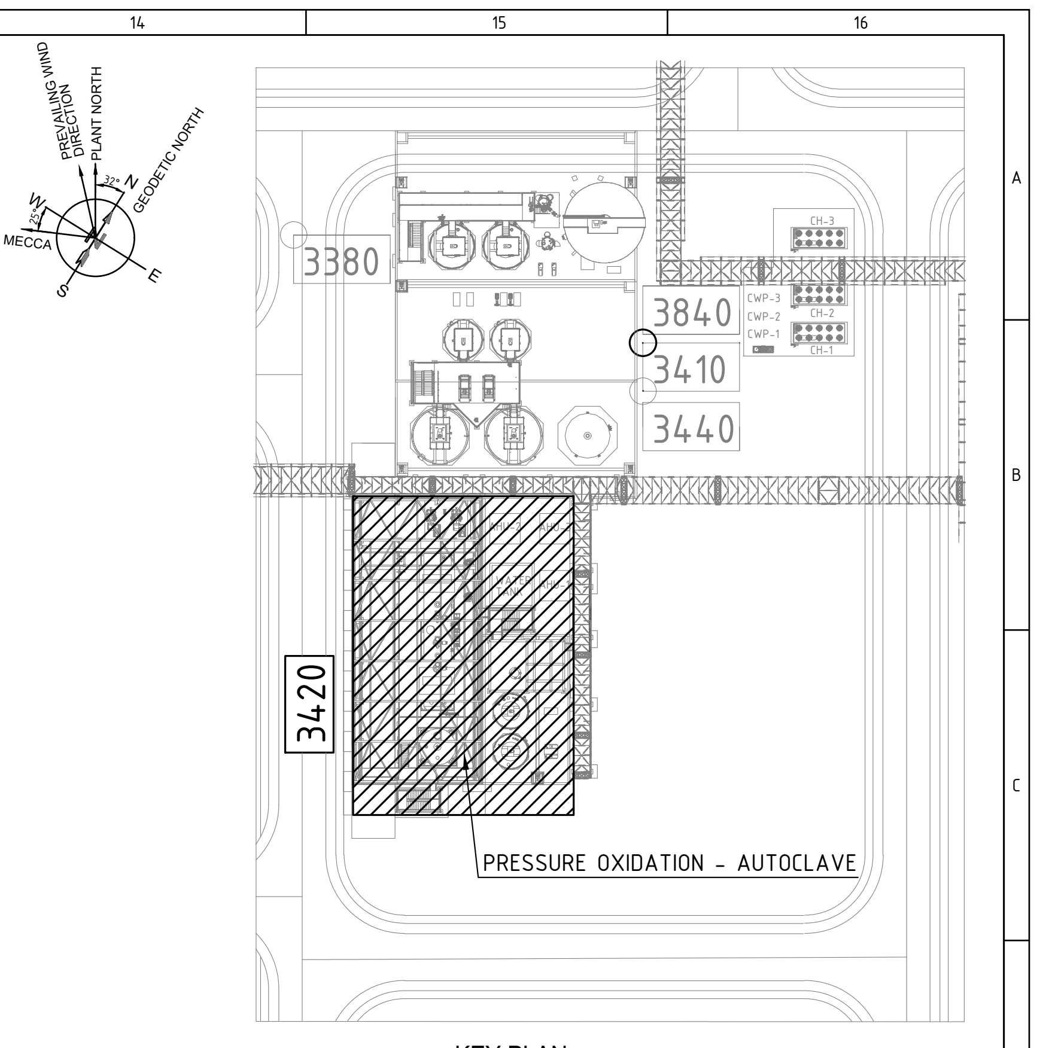
DETAIL OF P1,P1A,P1B,P2,P2A,P2B,
P3,P3A,P3B,P3C,P3D,P3E
SCALE 1:30



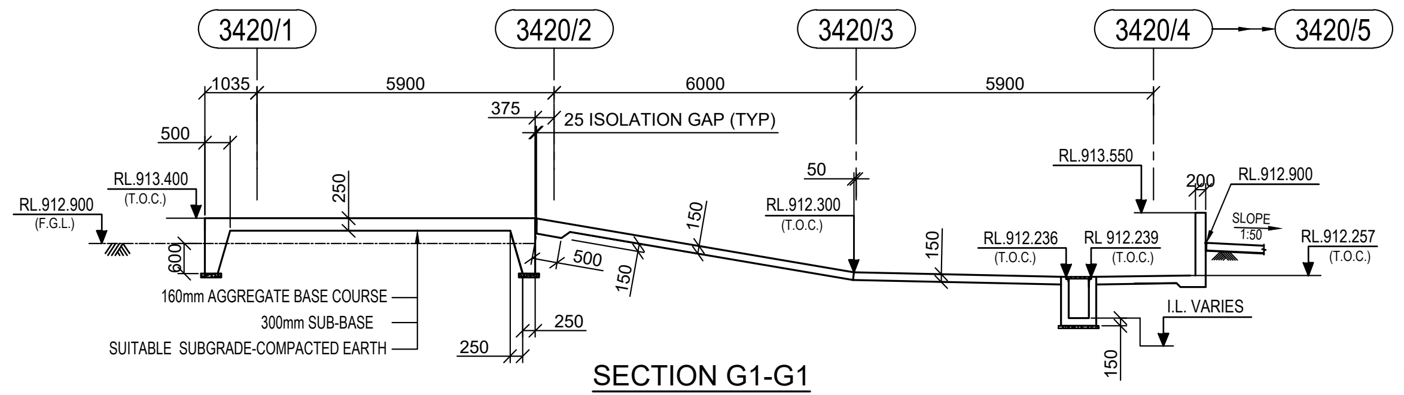
DETAIL OF P4, P5, P4A, P4B
SCALE 1:30



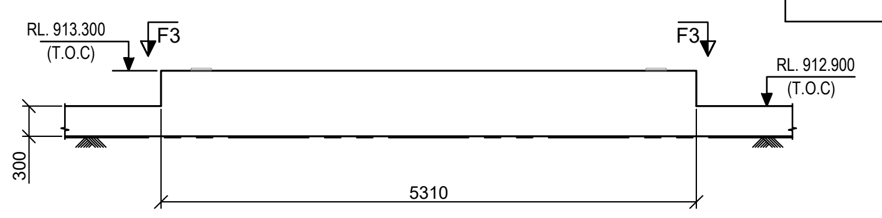
SECTION F3-F3
SCALE 1:50



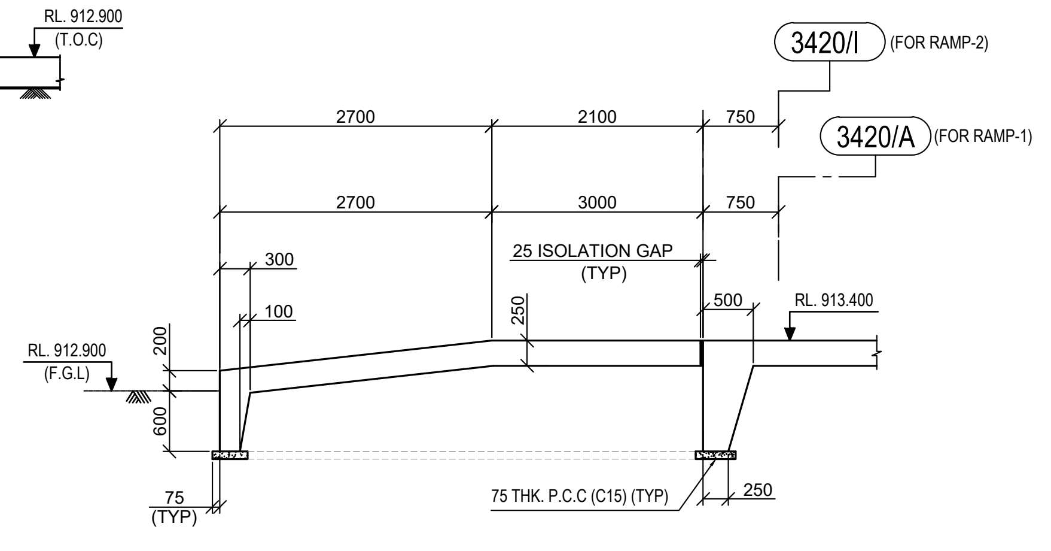
KEY PLAN
(SCALE- 1:750)



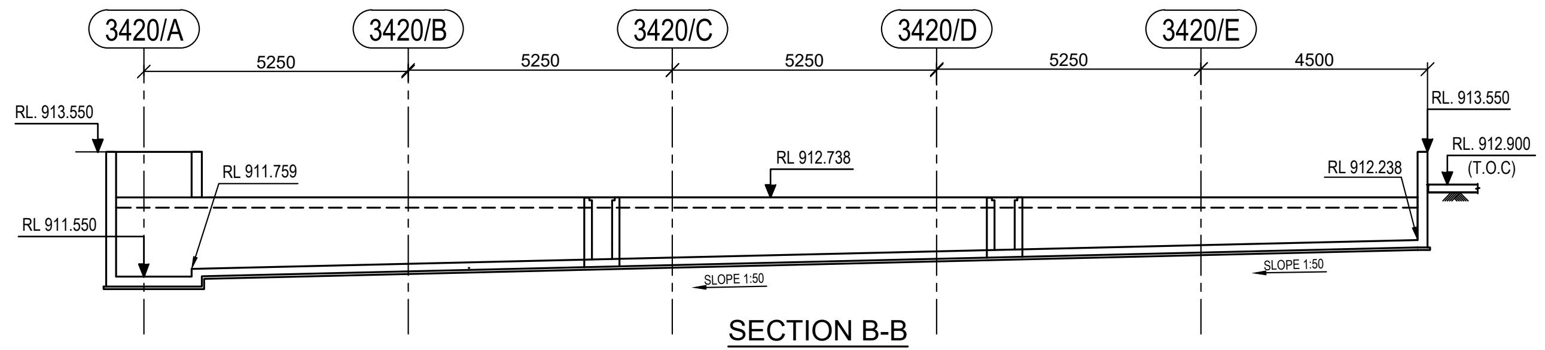
SECTION G1-G1
SCALE 1:100



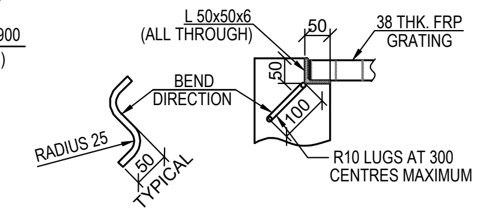
DETAIL OF P6
SCALE 1:50



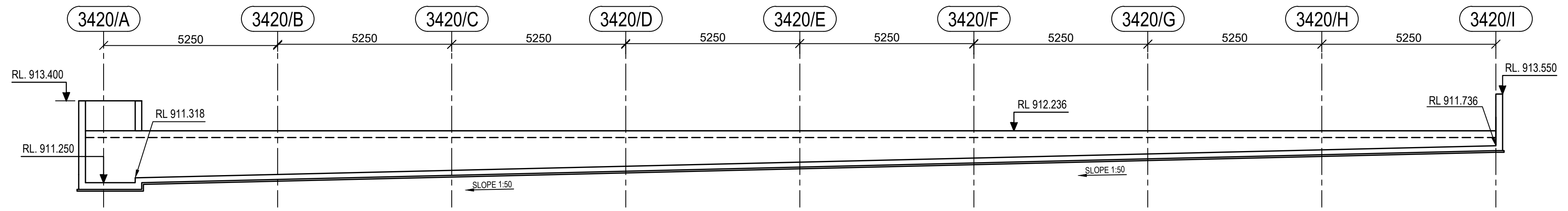
TYP DETAIL OF RAMP 1 & 2
SCALE 1:50



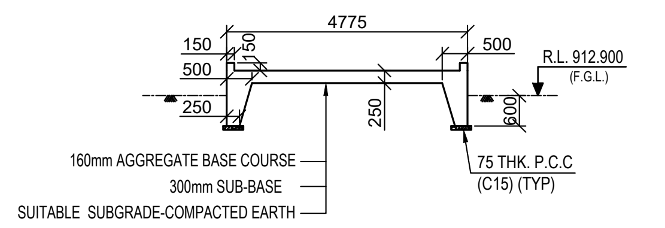
SECTION B-B
SCALE 1:100



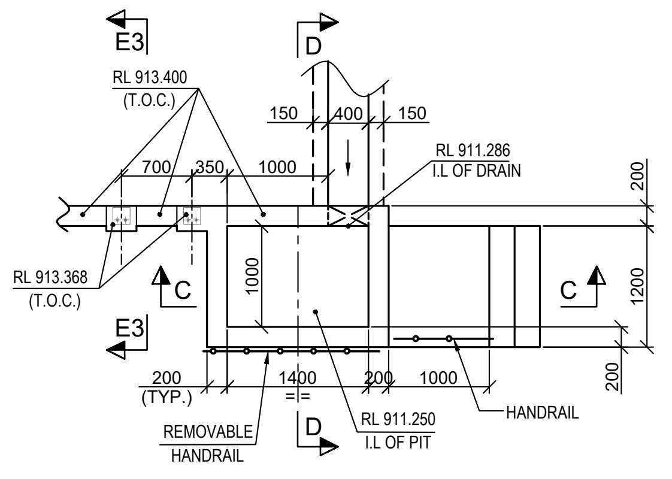
DETAIL - 'D3'
(SCALE 1:10)
(TYPICAL)



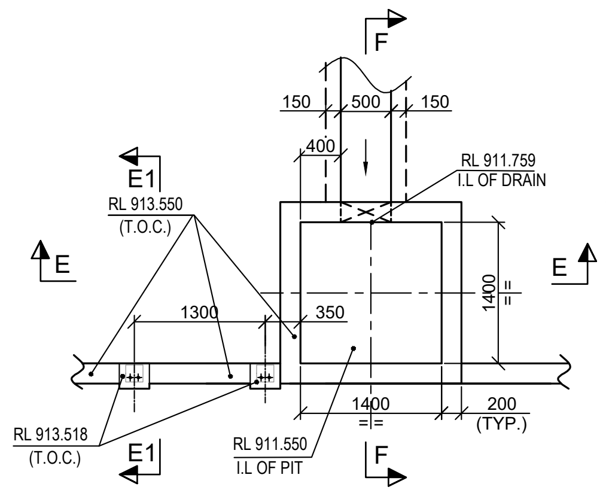
SECTION A-A
SCALE 1:100


160mm AGGREGATE BASE COURSE
300mm SUB-BASE
SUITABLE SUBGRADE-COMPACTED EARTH
75 THK. P.C.C (C15) (TYP)

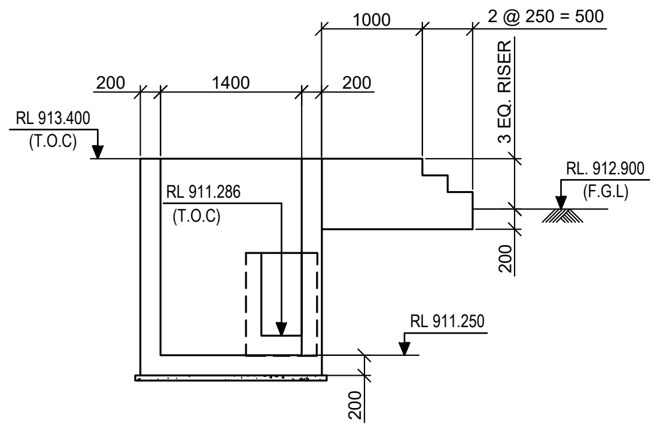
SECTION G2-G2
SCALE 1:100



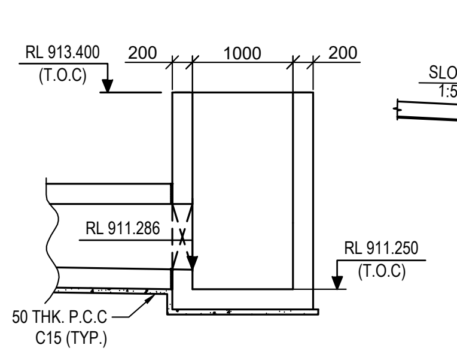
DETAIL-D1
SCALE 1:50



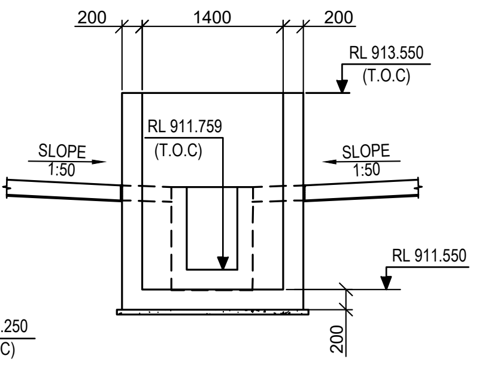
DETAIL-D2
SCALE 1:50



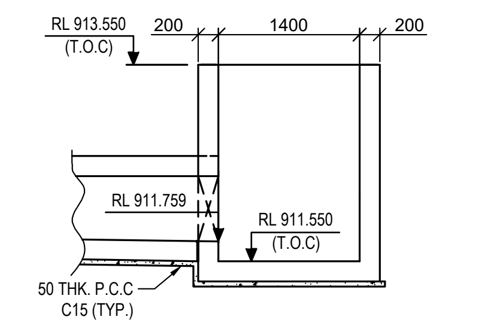
SECTION C-C
SCALE 1:50



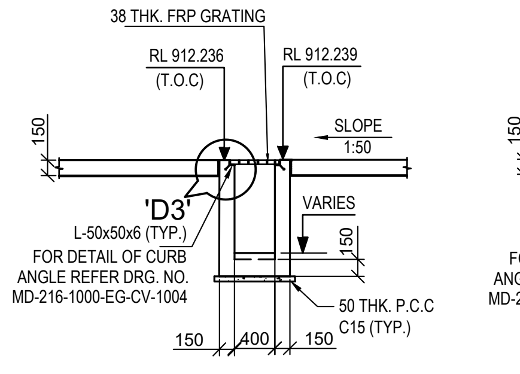
SECTION D-D
SCALE 1:50



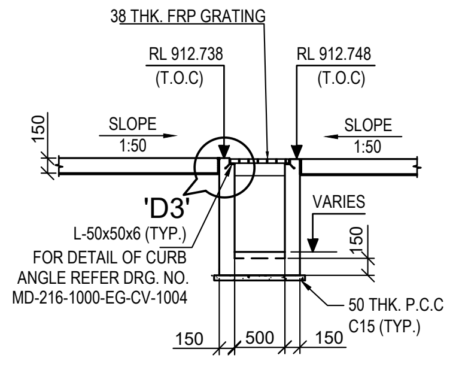
SECTION E-E
SCALE 1:50



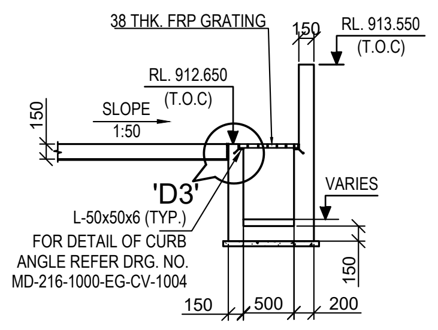
SECTION F-F
SCALE 1:50



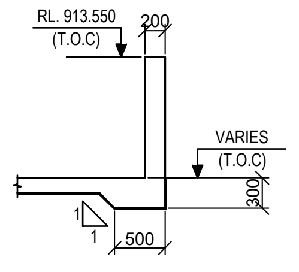
SECTION G-G
SCALE 1:50



SECTION H-H
SCALE 1:50



SECTION H1-H1
SCALE 1:50



SECTION J-J
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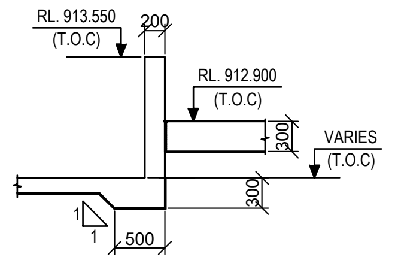
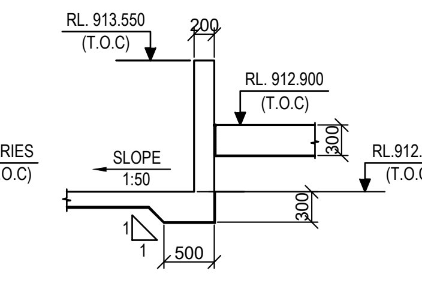
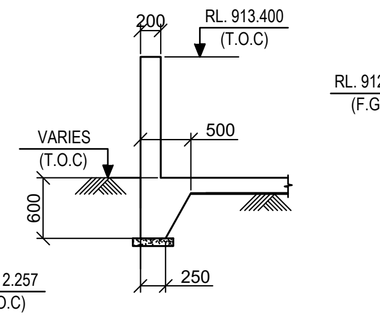


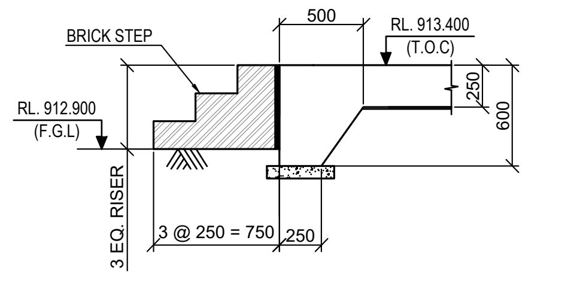
SECTION J1-J1
SCALE 1:50



SECTION K-K
SCALE 1:50



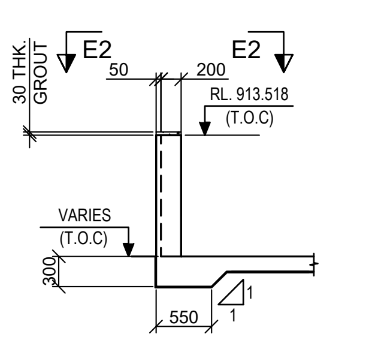
SECTION L-L
SCALE 1:50



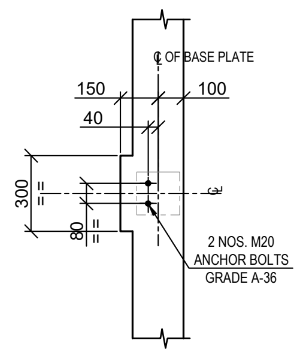
SECTION L1-L1
SCALE 1:30

LEGEND :-

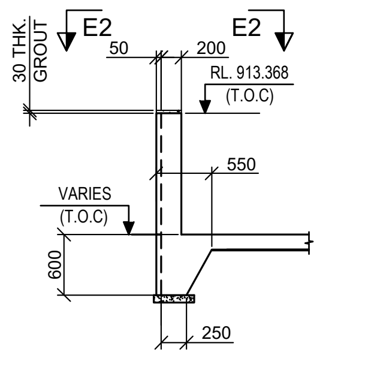
R.L. - REDUCED LEVEL
I.L - INVERT LEVEL
THK. - THICK
TYP. - TYPICAL.
T.O.C - TOP OF CONCRETE
₵ - CENTRE LINE
P.C.C - PLAIN CEMENT CONCRETE
DIA - DIAMETER
FRP - FIBRE REINFORCED POLYMER
F.G.L. - FINISHED GROUND LEVEL.
FDN. - FOUNDATION
AHU - AIR HANDLING UNIT
Ø - DIAMETER
R - RADIUS



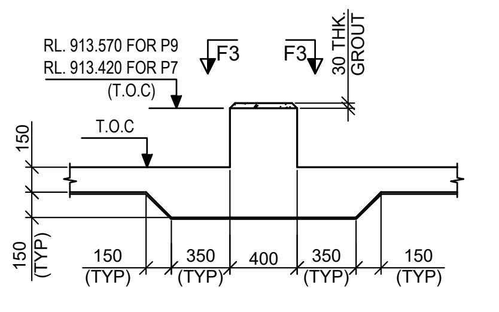
SECTION E1-E1
SCALE 1:50



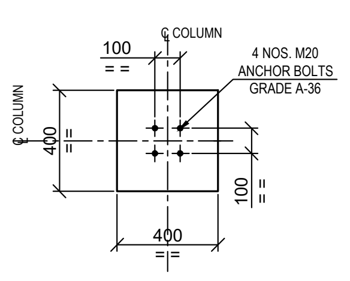
VIEW : E2-E2
SCALE 1:20



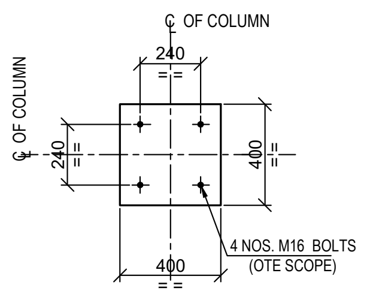
SECTION E3-E3
SCALE 1:50



DETAIL OF P7, P9
SCALE 1:30



VIEW : F3-F3
(FOR P7)
SCALE 1:20



VIEW : F3-F3
(FOR P9)
SCALE 1:20

NOTES:-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES UNLESS OTHERWISE STATED.
2. FOR DETAILS OF GRADE SLAB REFER DRAWING NO. MD-216-1000-EG-CV-DED-1002.
3. T.O.C. LEVEL DYKE WALLS SHALL BE MAINTAINED AT UNIFORM LEVEL WITH SECOND STAGE CONCRETE AS REQUIRED.

SHEET 1 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	15/03/2023	AS-BUILT	PML	SJM	SOR	AG	
		C05	15/03/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C05	PML	SJM	SOR	AG	
MD-216-3400-EG-EL-LAD-1006	PRESSURE OXIDATION	C04	26/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C04	PML	SJM	SOR	AG	
	CABLE ROUTE LAYOUT	C03	21/01/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C03	PML	SJM	SOR	AG	
MD-216-3420-EG-ST-GAD-2248	PRESSURE OXIDATION AREA - DUCT SUPPORTS	C02	04/01/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C02	PML	SJM	SOR	AG	
MD-216-3420-EG-ME-GAD-1011	PRESSURE OXIDATION - FEED STORAGE	C01	21/11/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C01	PML	SJM	SOR	AG	
	PLAN AT EL. +912.900	A01	20/10/2020	ISSUED FOR REFERENCE	PML	SJM	SOR	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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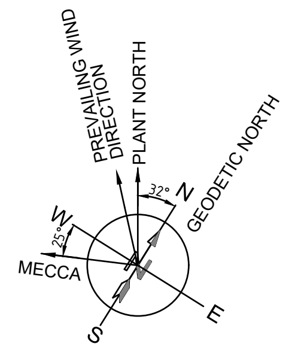
LARSEN & TOUBRO DRAWING NUMBER	019045_450_C_G_24566_001	REVISION 6
OUTOTEC DRAWING NUMBER	900707 __GAU01 __HH12 __MTL01 __34250	DOCUMENT ID 0U100704172 — REVISION 6

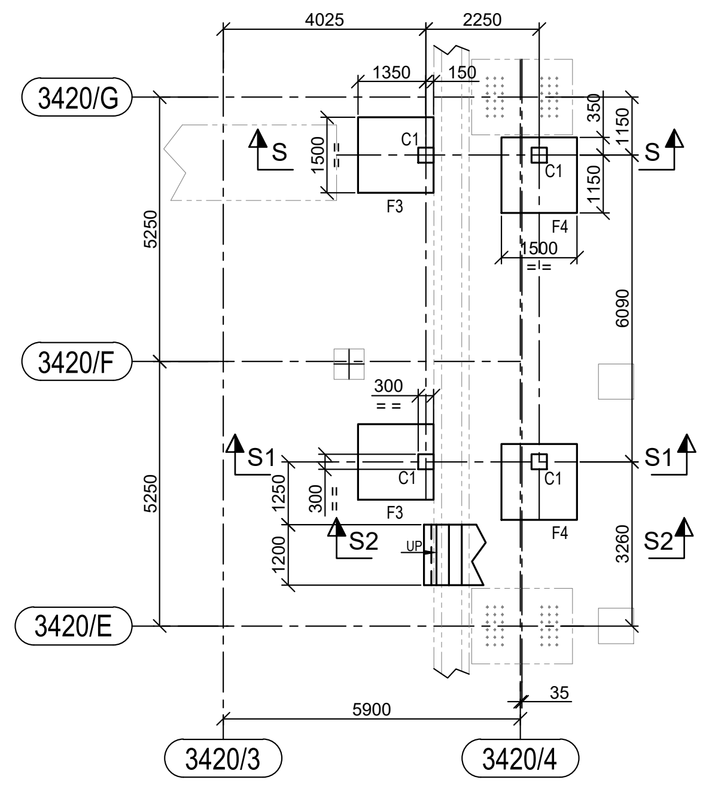
DRAWN	PML	14/10/2020
CHECKED	SJM	16/10/2020
ENGINEER	SOR	14/10/2020
APPROVED	AG	20/10/2020
EMPLOYER		
SCALE	1:100, 1:50, 1:25, 1:20	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRESSURE OXIDATION - AUTOCLAVE BUILDING
LAYOUT & DETAIL OF GRADE SLAB

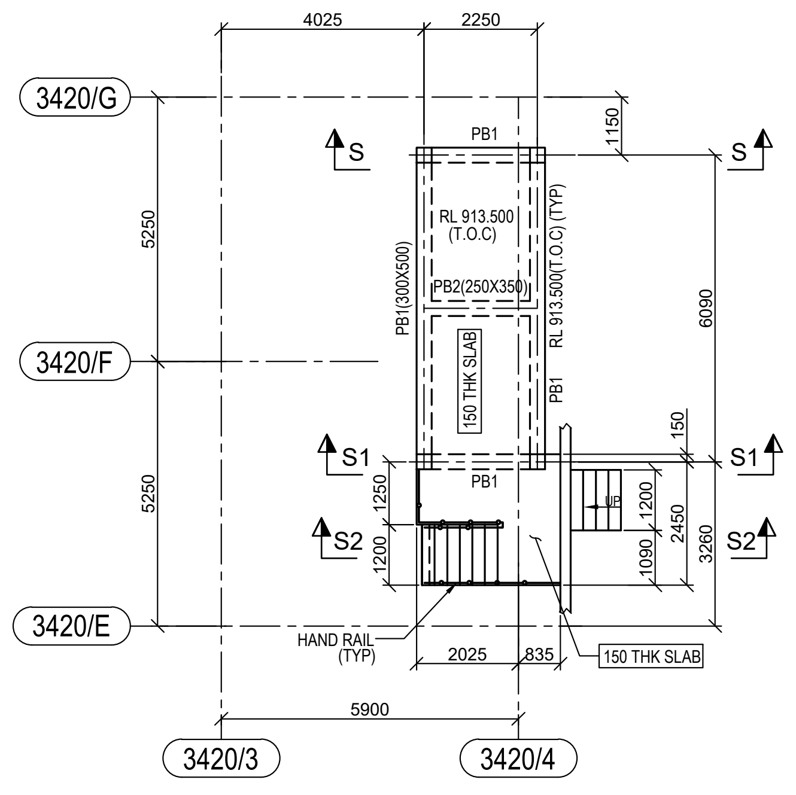
DRAWING NUMBER MD - 216 - 3420 - EG - CV - GAD - 1245
REV D01

DRAWING NUMBER
MD - 216 - 3420 - EG - CV - GAD - 1245
REV
D01

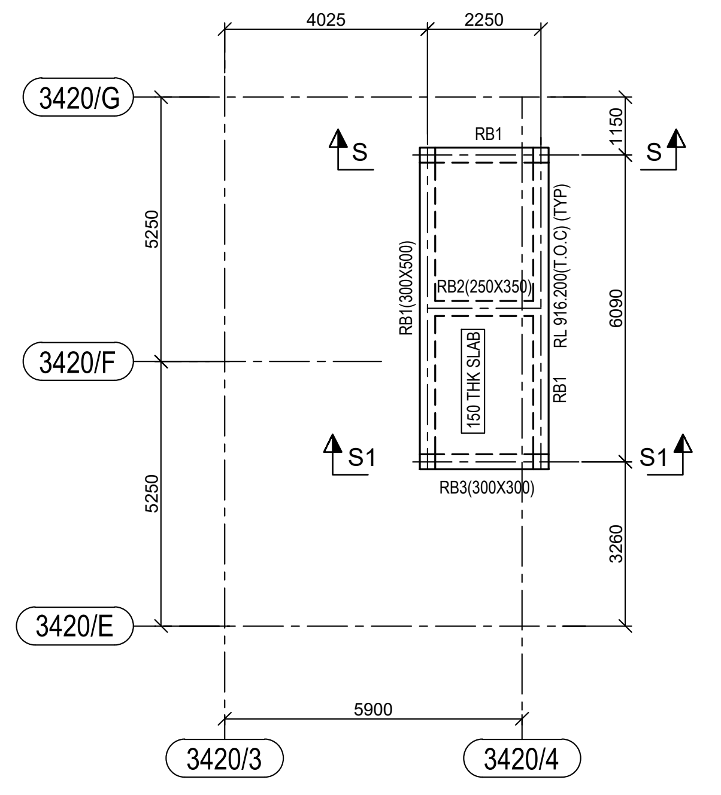




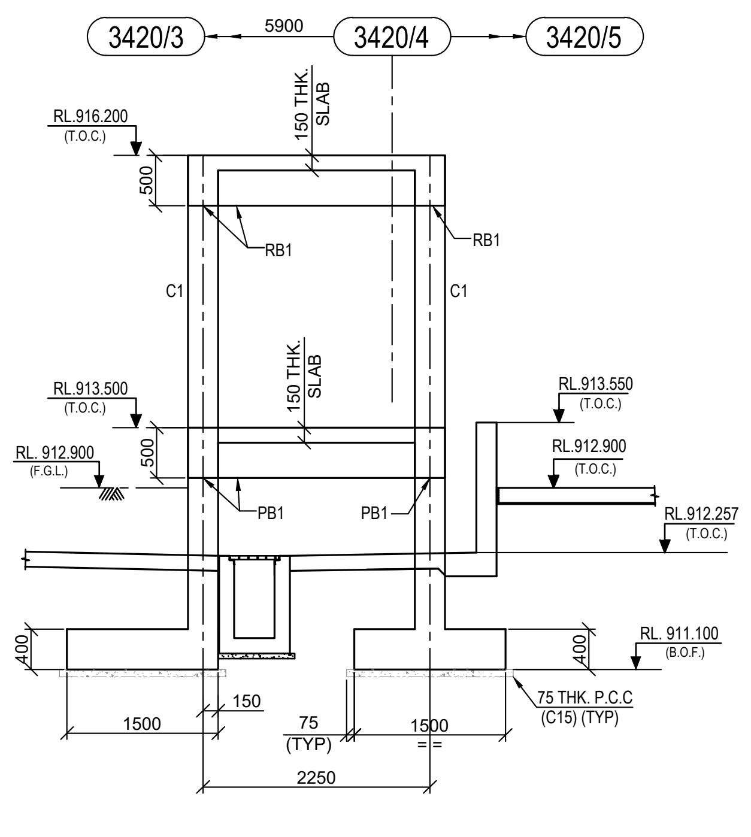
FOUNDATION PLAN OF AUTOCLAVE ROOM
(SCALE - 1:100)



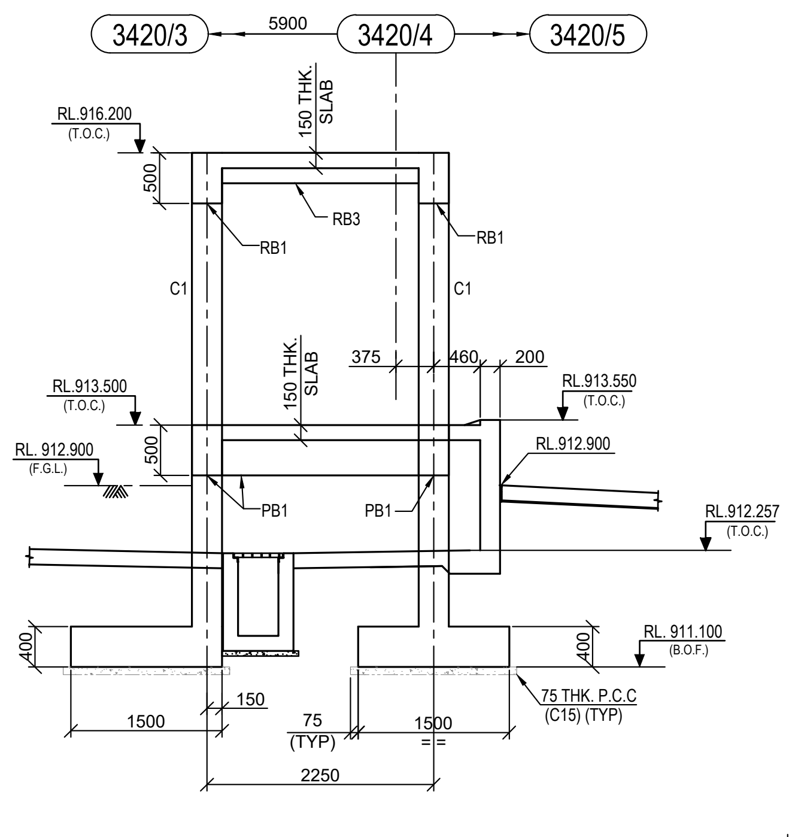
PLINTH BEAM PLAN OF AUTOCLAVE ROOM
(SCALE - 1:100)



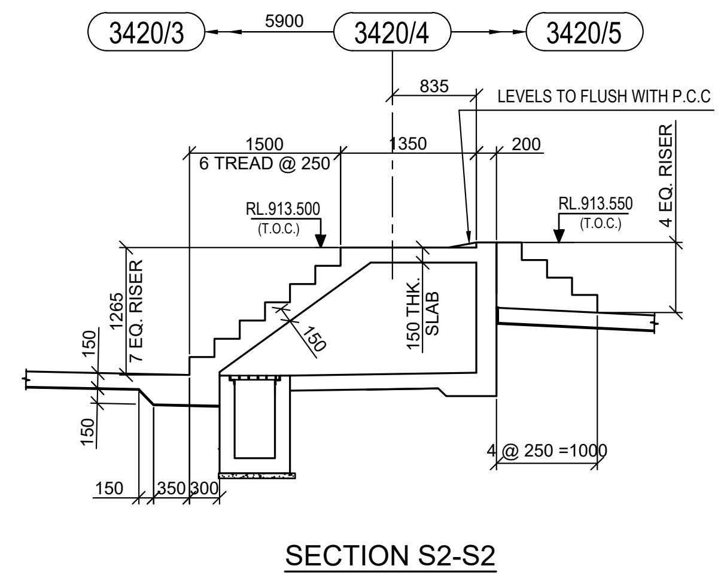
ROOF BEAM PLAN OF AUTOCLAVE ROOM
(SCALE - 1:100)



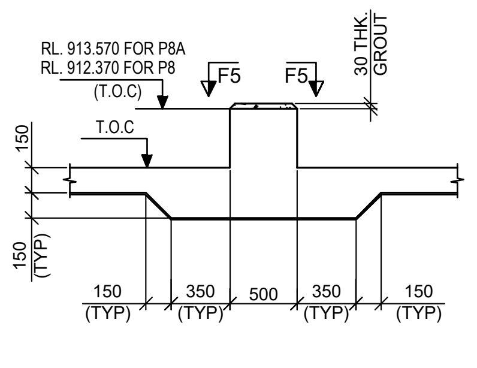
SECTION S-S
SCALE 1:50



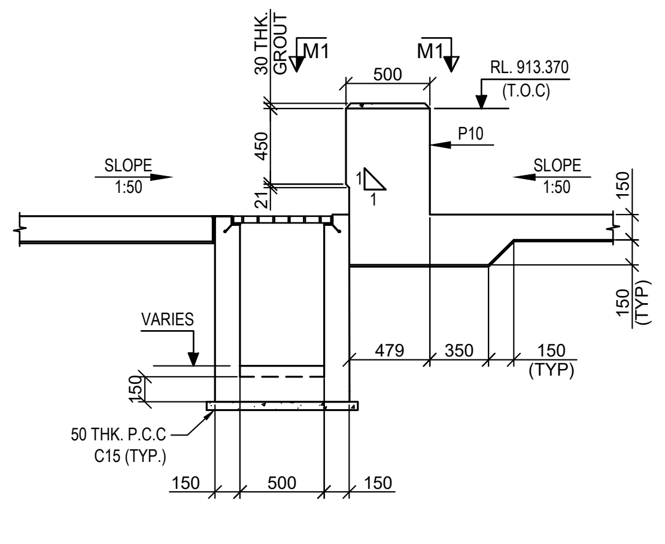
SECTION S1-S1
SCALE 1:50



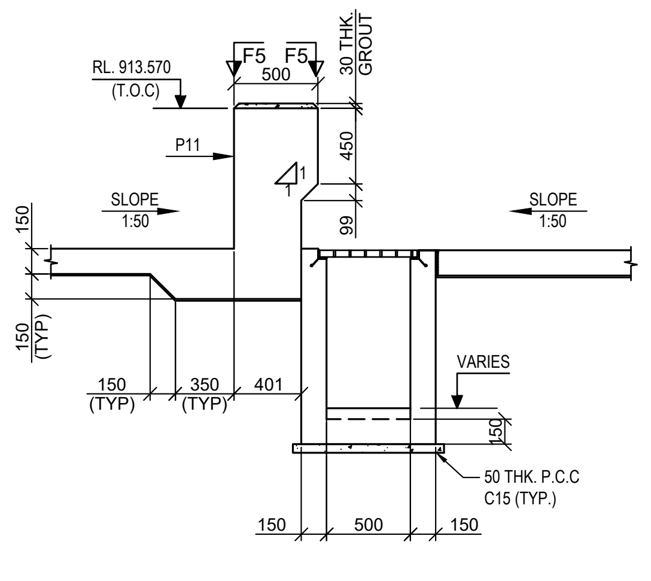
SECTION S2-S2
SCALE 1:50



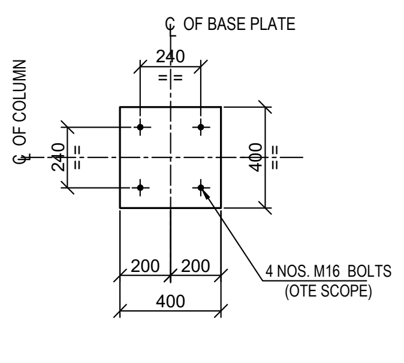
DETAIL OF P8,P8A
SCALE 1:30



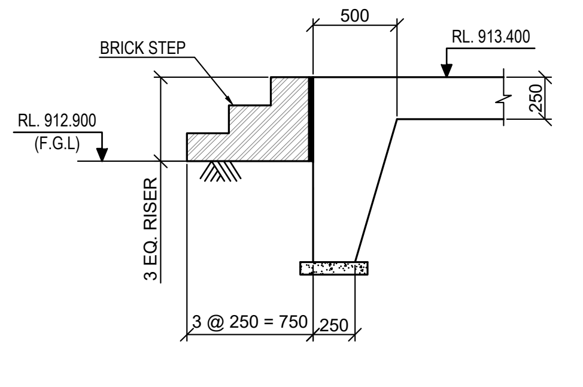
SECTION M-M
SCALE 1:30



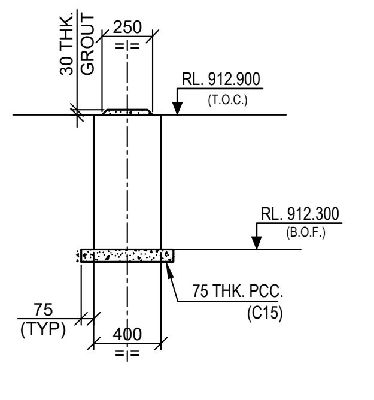
SECTION N-N
SCALE 1:30



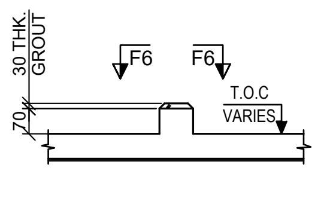
VIEW : M3-M3
SCALE 1:20



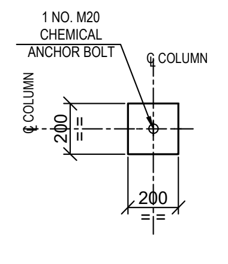
SECTION G3-G3
SCALE 1:30



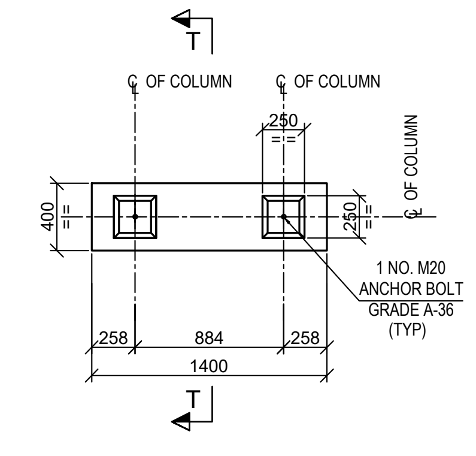
SECTION T-T
SCALE 1:30



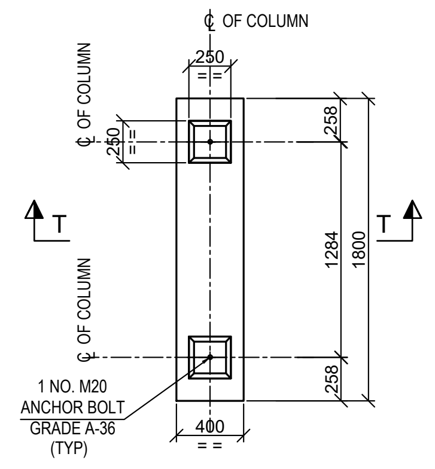
DETAIL OF P5A
SCALE 1:30



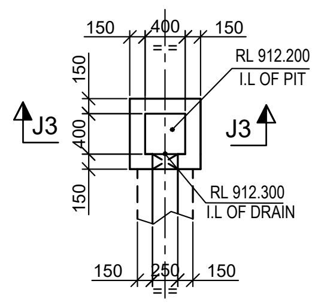
VIEW : F6-F6
SCALE 1:20



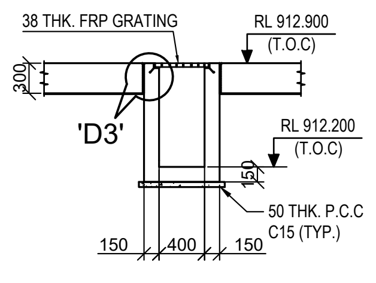
DETAIL OF FOUNDATION MKD F1
(2 NOS REQUIRED)
SCALE 1:30



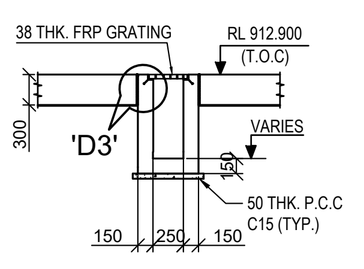
DETAIL OF FOUNDATION MKD F2
(1 NO REQUIRED)
SCALE 1:30



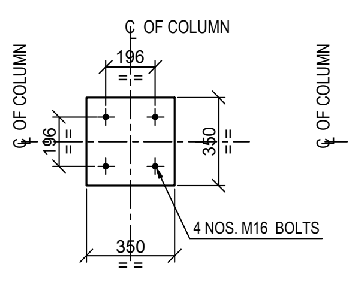
DETAIL-D4
SCALE 1:50



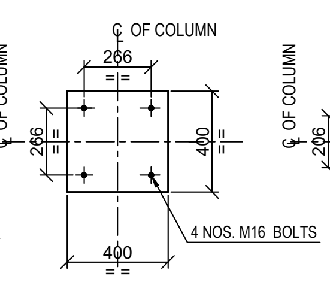
SECTION J3-J3
SCALE 1:50



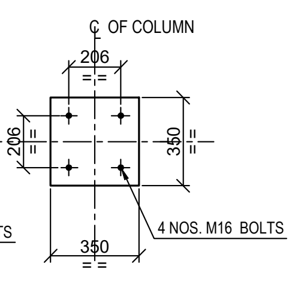
SECTION J2-J2
SCALE 1:50



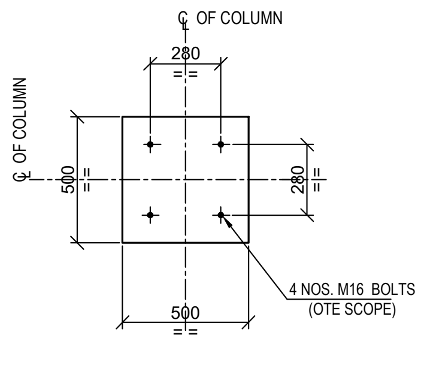
VIEW : F7-F7
(FOR P15)
SCALE 1:20



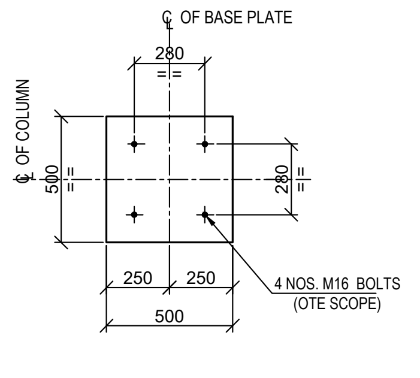
VIEW : F7-F7
(FOR P16)
SCALE 1:20



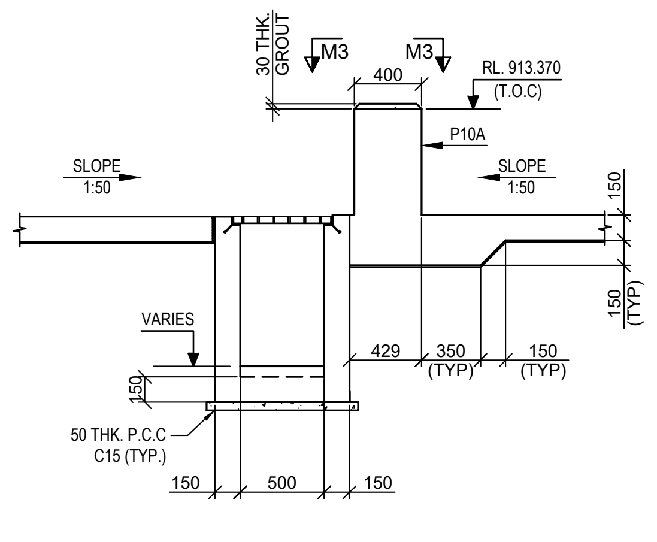
VIEW : F7-F7
(FOR P17,P18)
SCALE 1:20



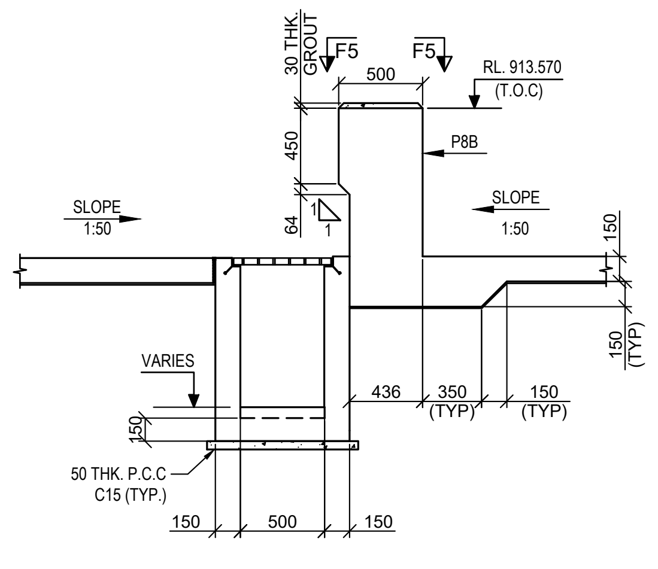
VIEW : F5-F5
SCALE 1:20



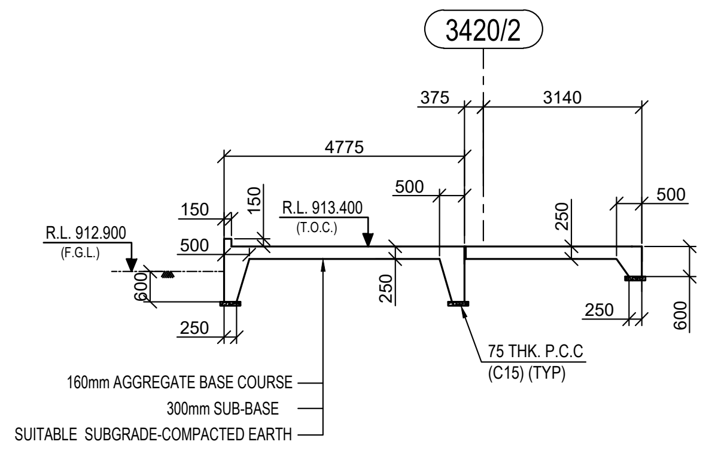
VIEW : M1-M1
SCALE 1:20



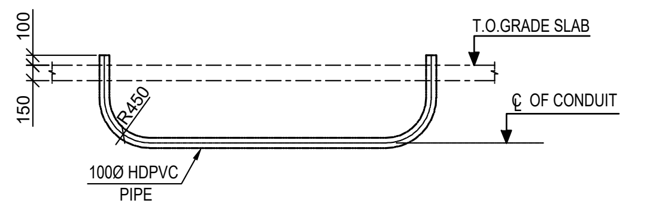
SECTION M2-M2
SCALE 1:30



SECTION P-P
SCALE 1:30

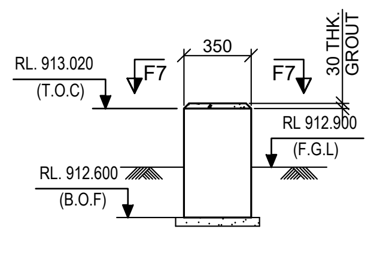

3420/2
375
3140
4775
RL 912.900 (F.G.L)
RL 913.400 (T.O.C)
75 THK. P.C.C (C15) (TYP)
160mm AGGREGATE BASE COURSE
300mm SUB-BASE
SUITABLE SUBGRADE-COMPACTED EARTH

SECTION G4-G4
SCALE 1:100

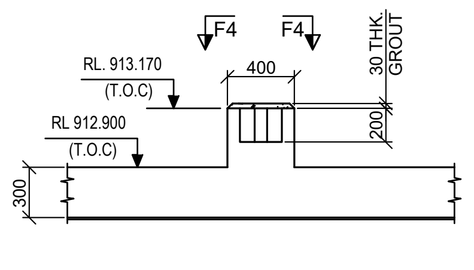


TYPICAL DETAIL OF CONDUIT
SCALE 1:50

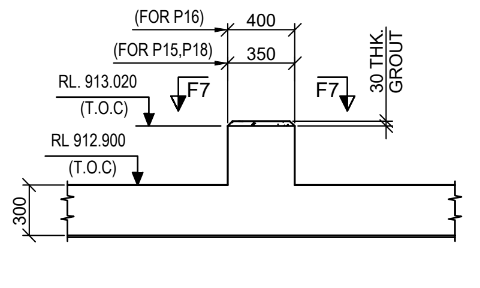
CONDUIT MKD.	ELEVATION AT ℄ OF CONDUIT
LV_HDPVC_001	RL 912.200
LV_HDPVC_002	RL 912.200
LV_HDPVC_003	RL 912.070
LV_HDPVC_004	RL 912.070
LV_HDPVC_005	RL 912.400
LV_HDPVC_006	RL 912.400
LV_HDPVC_018	RL 911.800
LV_HDPVC_019	RL 911.800
LV_HDPVC_020	RL 911.800
LV_HDPVC_021	RL 911.800
LV_HDPVC_022	RL 911.800
LV_HDPVC_023	RL 911.800
LV_HDPVC_032	RL 912.050
LV_HDPVC_033	RL 912.050



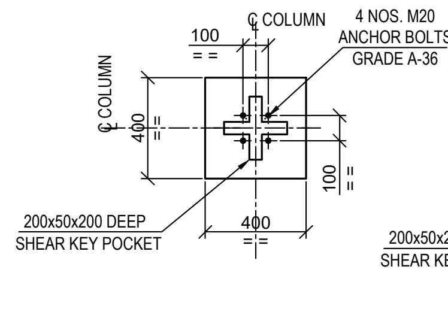
DETAIL OF P17
SCALE 1:30



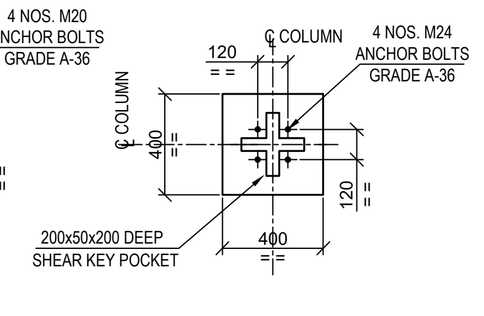
DETAIL OF P13 & P14
SCALE 1:30



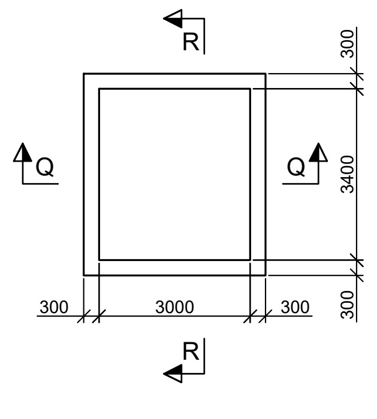
DETAIL OF P15, P16 & P18
SCALE 1:30


COLUMN
4 NOS. M20 ANCHOR BOLTS GRADE A-36
200x50x200 DEEP SHEAR KEY POCKET
400

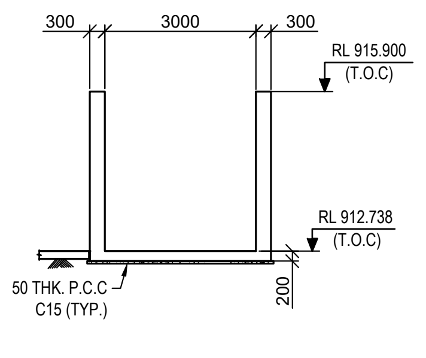
VIEW : F4-F4
(FOR P13)
SCALE 1:20


COLUMN
4 NOS. M24 ANCHOR BOLTS GRADE A-36
200x50x200 DEEP SHEAR KEY POCKET
400

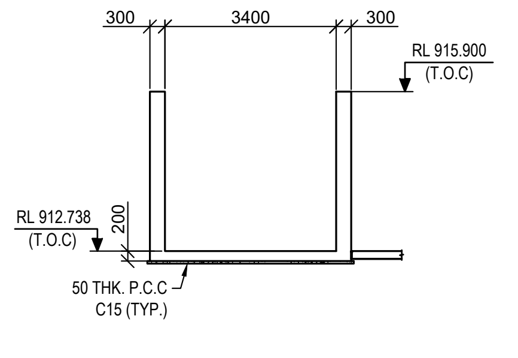
VIEW : F4-F4
(FOR P14)
SCALE 1:20



DETAIL OF BUNKER
SCALE 1:100



SECTION Q-Q
SCALE 1:100



SECTION R-R
SCALE 1:100

LEGEND :-

R.L. - REDUCED LEVEL
I.L - INVERT LEVEL
THK. - THICK
TYP. - TYPICAL.
MKD. - MARKED
T.O.C - TOP OF CONCRETE
T.O.G - TOP OF GROUT
℄ - CENTRE LINE
P.C.C - PLAIN CEMENT CONCRETE
F.G.L. - FINISHED GROUND LEVEL.
GA - GENERAL ARRANGEMENT
B.O.F. - BOTTOM OF FOUNDATION
DIA - DIAMETER
CJ - CONSTRUCTION JOINT
FDN. - FOUNDATION
STR. - STRUCTURAL
EQ. - EQUAL
REQD. - REQUIRED
RCC - REINFORCED CEMENT CONCRETE
Ø - DIAMETER
R - RADIUS

NOTES:-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) FOR FOUNDATION & COLUMN.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
5. FOR STANDARD REINFORCEMENT DETAIL & GENERAL NOTES FOR CIVIL WORKS REFER DRAWING NO. MD-216-1000-EG-CV-DED-1005 &1001.
6. FOR REINFORCEMENTS DETAIL REF. DRG. NO. MD - 216 - 3420 - EG - CV - GAD - 1246.

SHEET 2 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	15/03/2023	AS-BUILT	PML	SJM	SOR	AG	
		C05	15/03/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C05	PML	SJM	SOR	AG	
		C04	26/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C04	PML	SJM	SOR	AG	
MD-216-3400-EG-EL-LAD-1006	PRESSURE OXIDATION	C03	21/01/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C03	PML	SJM	SOR	AG	
	CABLE ROUTE LAYOUT	C02	04/01/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C02	PML	SJM	SOR	AG	
MD-216-3420-EG-ME-GAD-1011	PRESSURE OXIDATION - FEED STORAGE	C01	21/11/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C01	PML	SJM	SOR	AG	
	PLAN AT EL. +912.900	A01	20/10/2020	ISSUED FOR REFERENCE	PML	SJM	SOR	AG	

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24566_001 — REVISION 6
OUTOTEC DRAWING NUMBER: 900707 __ GAU01 __ HH12 __ MTL01 __ 34250 — DOCUMENT ID: OU100704172 — REVISION 6

DRAWN	PML	14/10/2020
CHECKED	SJM	16/10/2020
ENGINEER	SOR	14/10/2020
APPROVED	AG	20/10/2020
EMPLOYER		
SCALE	1:100, 1:50, 1:25, 1:20	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRESSURE OXIDATION - AUTOCLAVE BUILDING
LAYOUT & DETAIL OF GRADE SLAB

DRAWING NUMBER: MD - 216 - 3420 - EG - CV - GAD - 1245
REV: D01